Distributed EV charging network

Hybrid MPI and OpenMP simulation

Three variants. 30 grid sizes. 30 line sizes. CAAS verification.

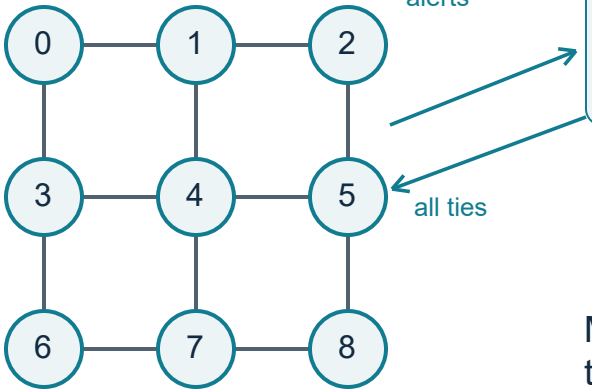
A measured study of neighbour communication and base-station congestion handling.

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AI use declaration: slide 13.

One station process, one thread per port



alerts

all ties

Base station: world rank 0

Station i: world rank i+1

OpenMP team (k=4 shown)

T0 **T1** **T2** **T3**

Shared ports: [1, 0, 1, 0]

MPI_THREAD_FUNNELED: only the initial thread calls MPI. Barriers publish port snapshots.

MPI Cartesian stations 0..N-1

No wraparound. Missing edges: MPI_PROC_NULL.

Congestion triggers neighbour queries, then alerts

```
broadcast CONTINUE
threads update ports[i]
barrier; count free ports
exchange query intents with neighbours
reply only to requesting neighbours
if free < threshold:
    if neighbour has capacity: redirect
    else: send packed alert; receive advice
```

Query intent: 1 integer per real edge.

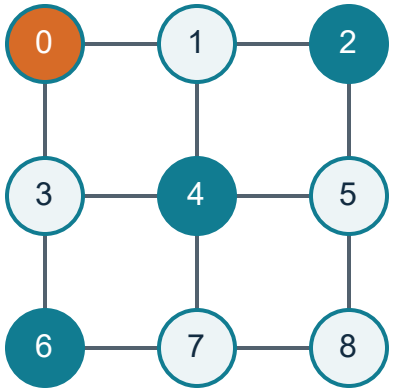
Report: ID, coordinates, UTC timestamp, capacities, counts and bytes.

Snapshot safety: one writer per port, atomic count, OpenMP barriers.

Tests: isolated node, full network, empty network and all ties.

Persistent threads avoid creating a new team for each simulation cycle.

The base returns every inferred nearest candidate



Orange: reporting station 0

Teal: equal-distance nodes 2, 4, 6

Receive the full batch, then update the cache.

Search the two-hop frontier in the Task 2 example. Manhattan distance = 2.

Exclude recent self-alerts and fresh low-capacity observations.

Return all remaining ties, or an empty list. Silence permits inferred availability.

Expiry is configurable. No reservation or guaranteed capacity is claimed.

Three versions preserve the same decisions

Version	Neighbour exchanges	Base communication	Candidate work
V0	Blocking Sendrecv	Rank-order Recv Blocking Send	Scan N nodes
V1	Irecv / Isend Waitall	Probe ANY_SOURCE Blocking Send	8 frontier offsets
V2	Irecv / Isend Waitall	Preposted Irecv Batched Isend	8 frontier offsets

Selection work per alert falls from O(N) to O(1). Waitall completes requests for safe buffer reuse.
Useful computation/communication overlap is not measured.

Timers measure local elapsed communication costs

Local study: two TCP-connected containers on one shared 16-CPU host.

30 grid sizes and 30 line sizes x 3 variants x 3 launches = 540 runs.

Two warmup cycles, six measured cycles, two port threads, mixed congestion.

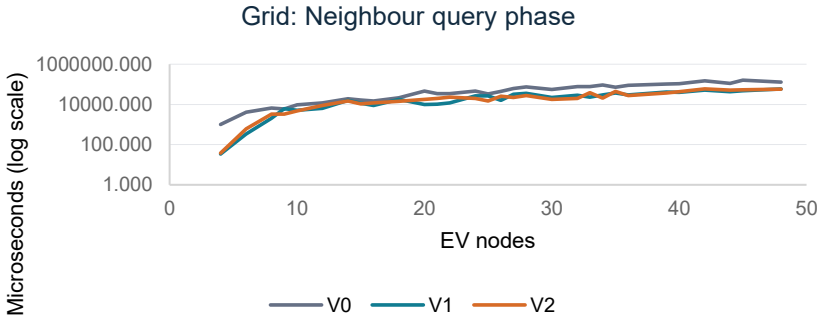
MPI_Wtime starts and stops on the same rank. No cross-host clock subtraction.

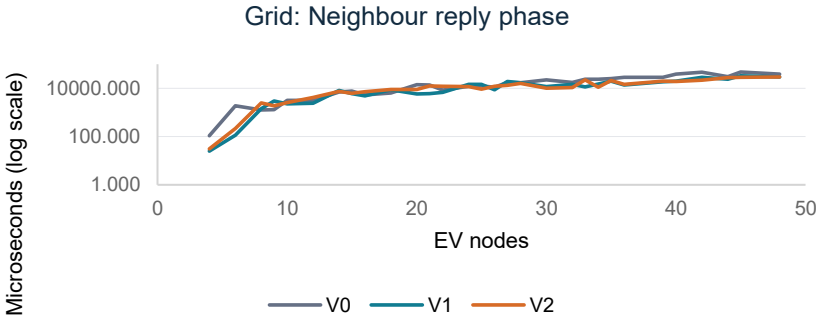
Query/reply: phase completion. Alert send: MPI completion. Redirect: receive wait.

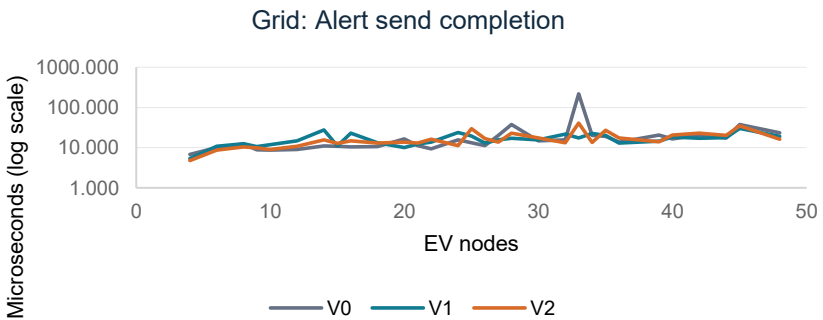
Round trip includes queueing and base computation. Raw rank-level records remain available.

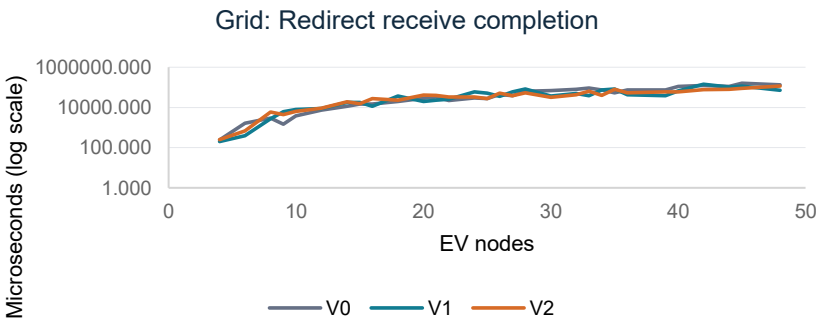
CAAS job 199603: 90 scaling runs, 5 grids + 5 lines, two allocated compute nodes.

Local grid scaling: bounded messages, increasing load



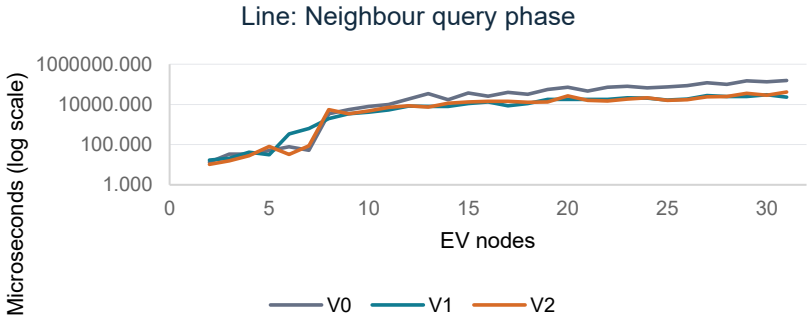


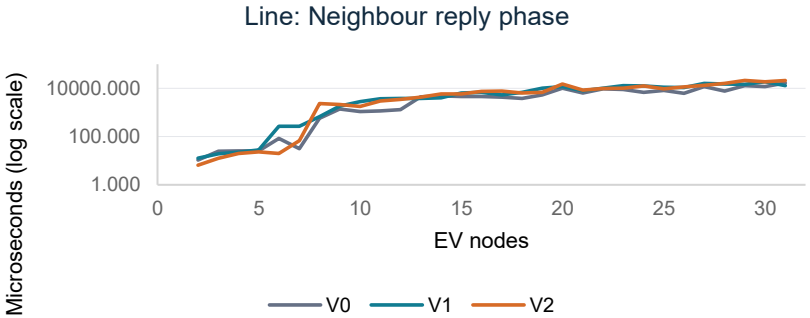


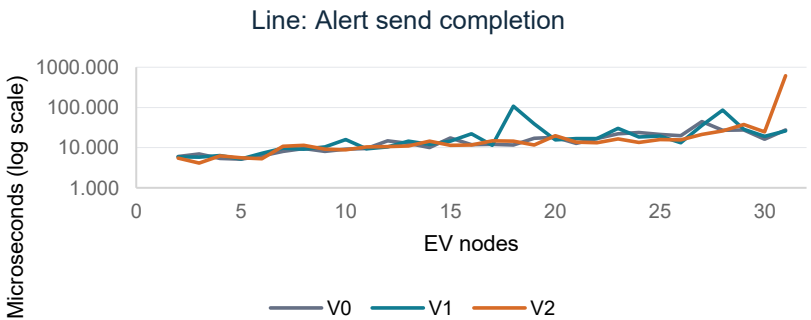


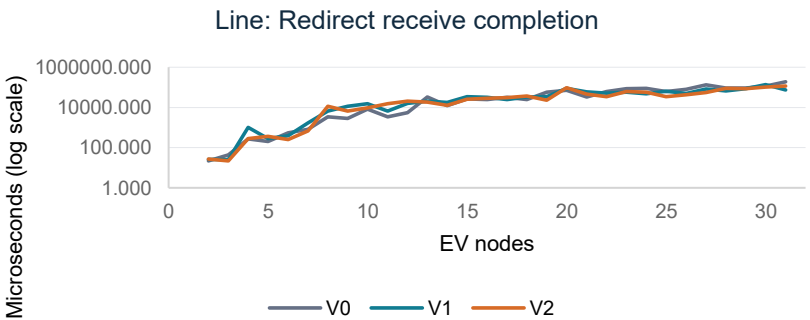
Selected V2, query: first five sizes 3312.2 us, last five 52526.6 us. Full ranges in appendix/CSV.

Local line scaling: bounded messages, increasing load





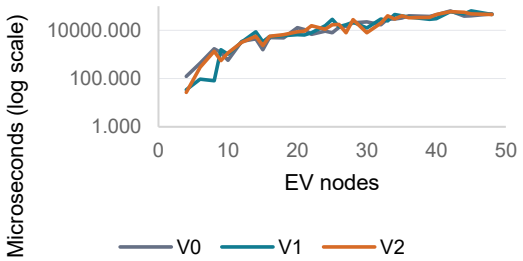




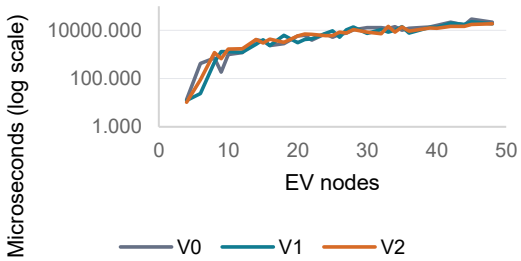
Selected V2, query: first five sizes 27.7 us, last five 28466.8 us. Full ranges in appendix/CSV.

Local control traffic and measured delay

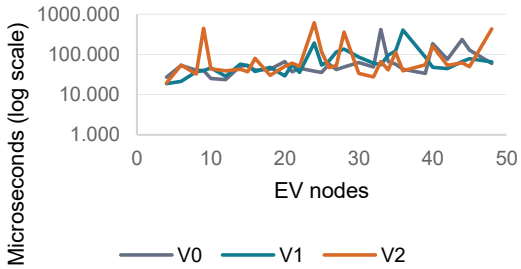
Grid: Control broadcast



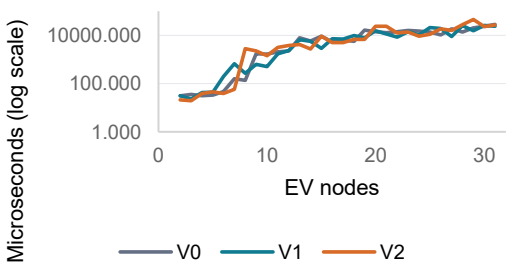
Grid: Alert-flag gather



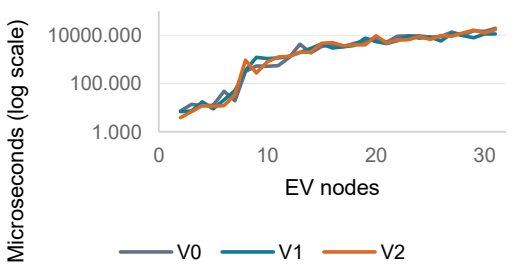
Grid: Shutdown ACK



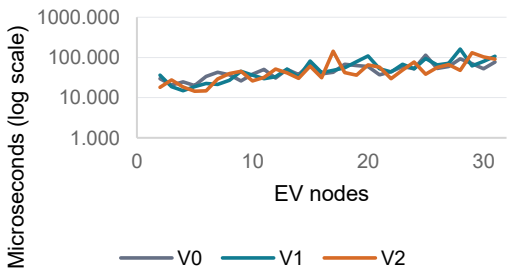
Line: Control broadcast



Line: Alert-flag gather

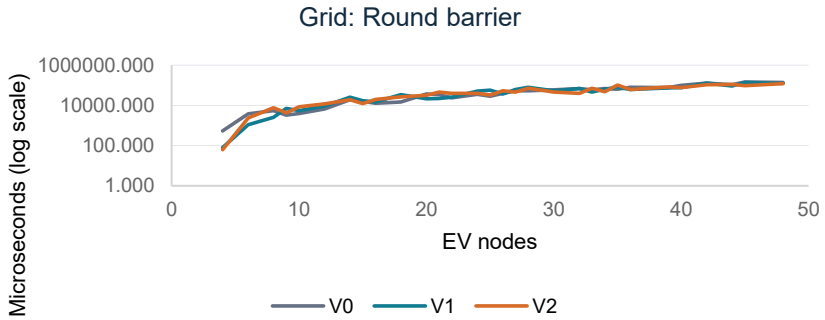


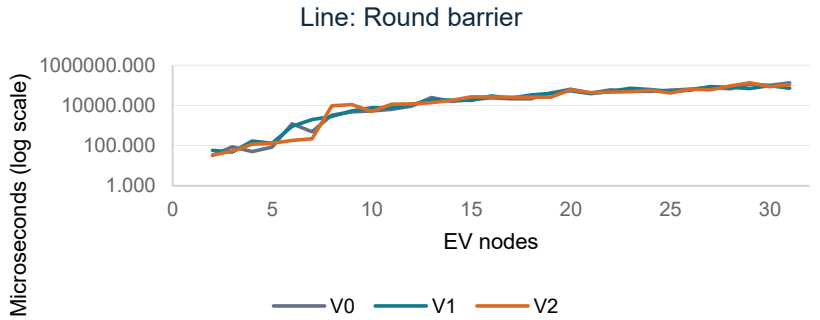
Line: Shutdown ACK

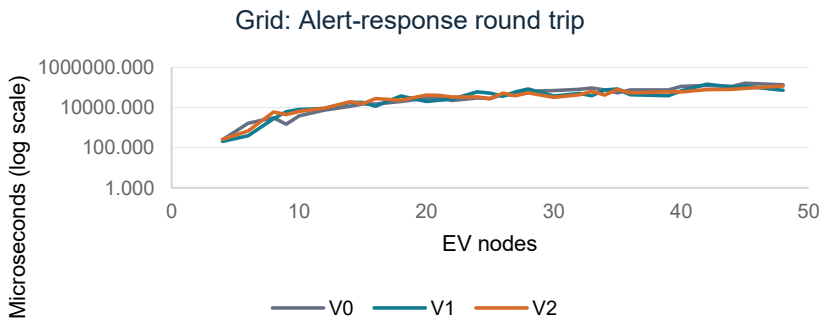


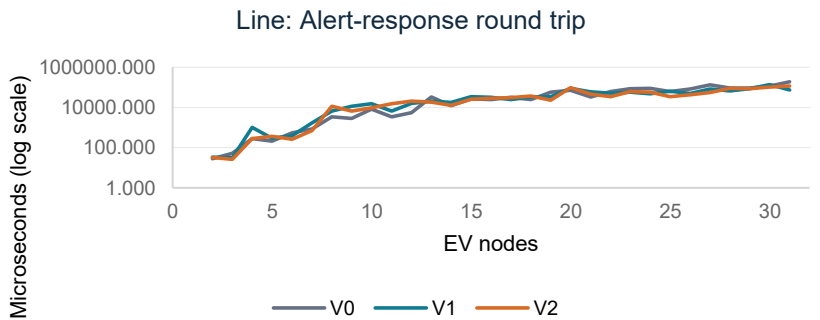
Broadcast/gather are collective elapsed times. ACK is sender completion. Every plotted category also has a detailed appendix graph.

Local synchronization and response times









Barrier: waiting for the round. Round trip: report send through advice receipt, including base processing.

CAAS measurements and implementation trade-offs

CAAS: 5 grid sizes + 5 lines x 3 variants x 3 launches = 90 scaling runs.

Grid query: V0/V1 = 1.64x; V0/V2 = 1.81x.

Line query: V0/V1 = 2.34x; V0/V2 = 2.49x.

Grid round trip: V0/V2 = 1.09x; line = 1.30x.

Highest observed round-trip score: V2 (1.197).

Ratios above 1 favour the refined version. Three launches; descriptive evidence.

Candidate compute at N=10,000: V0/V2 = 1709.9x (isolated).

Local and CAAS results are separate. No universal best variant is established.

Evidence expiry trades caution for recovery speed

CAAS expiry (s)	Alerts	Candidate recommendations
0	24	6
0.1	24	6
1	24	0
30	24	0

Six rounds; accelerated 0.05-second tick. Shorter expiry can forget congestion sooner. Missing evidence does not guarantee capacity.

Limitations: synthetic arrivals, global round barriers, inferred capacity and three repeats. Next: longer controlled runs, portable reports and candidate confirmation.

AI use declaration

Tool and scope

OpenAI Codex substantially assisted implementation and documentation: code review and fixes, validation, CAAS execution, results analysis, graphs and slides.

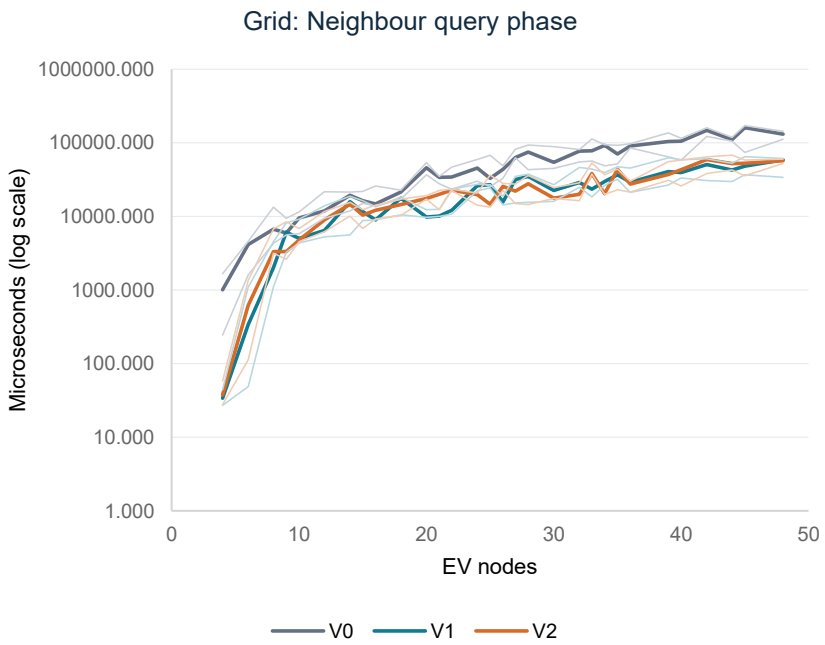
Student responsibility

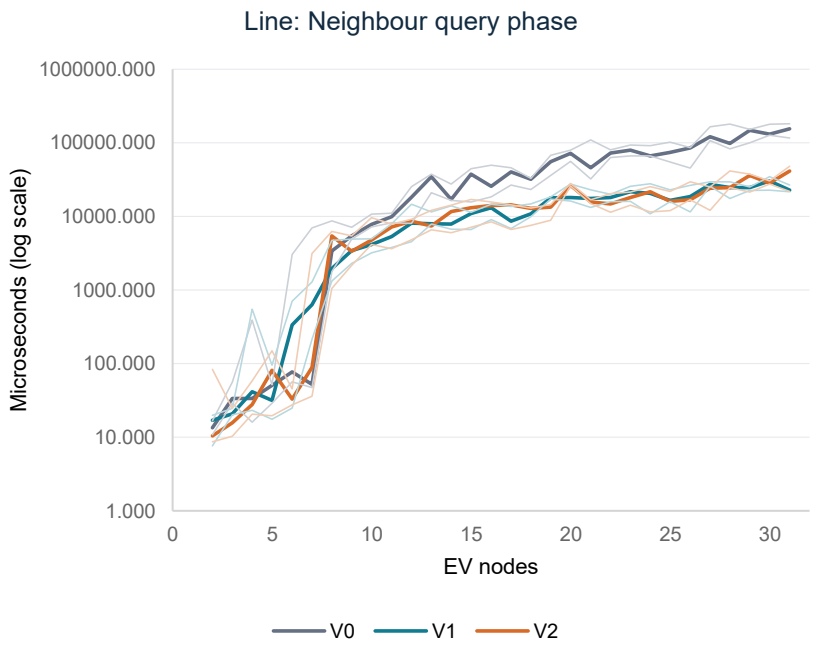
Both students are responsible for understanding and verifying the submitted code, measurements and claims. AI-generated material is not claimed as independent student authorship.

Preparation and assessment

AI assisted preparation. The students must deliver the assessed presentation and interview themselves, without AI.

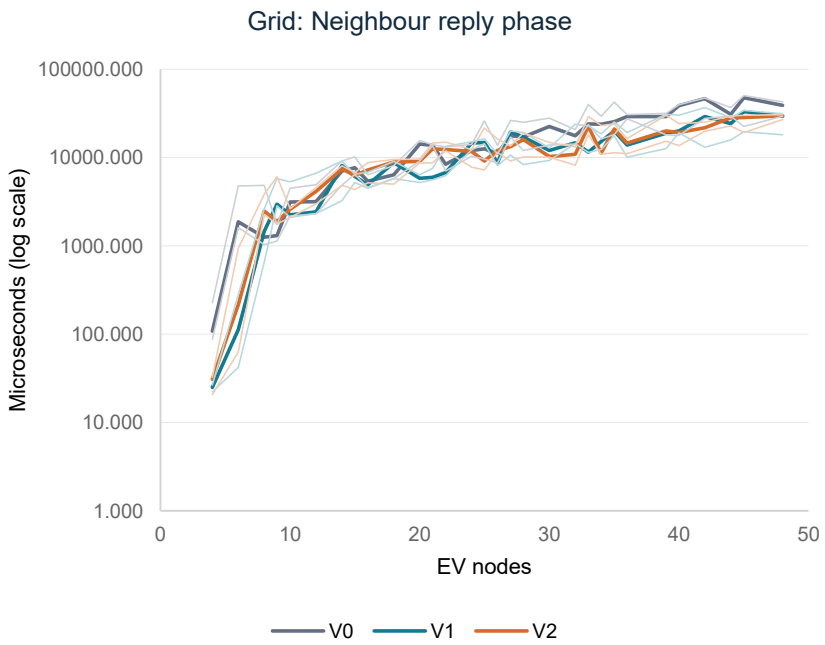
Neighbour query phase

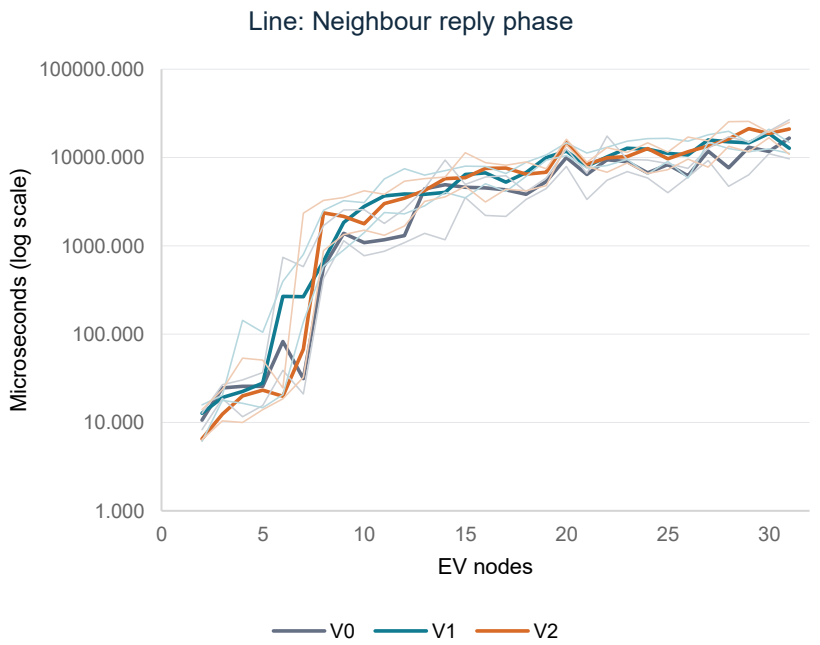




Every measured size is retained. Lines show medians of three independent run means. Pale boundary lines show the observed minimum and maximum.

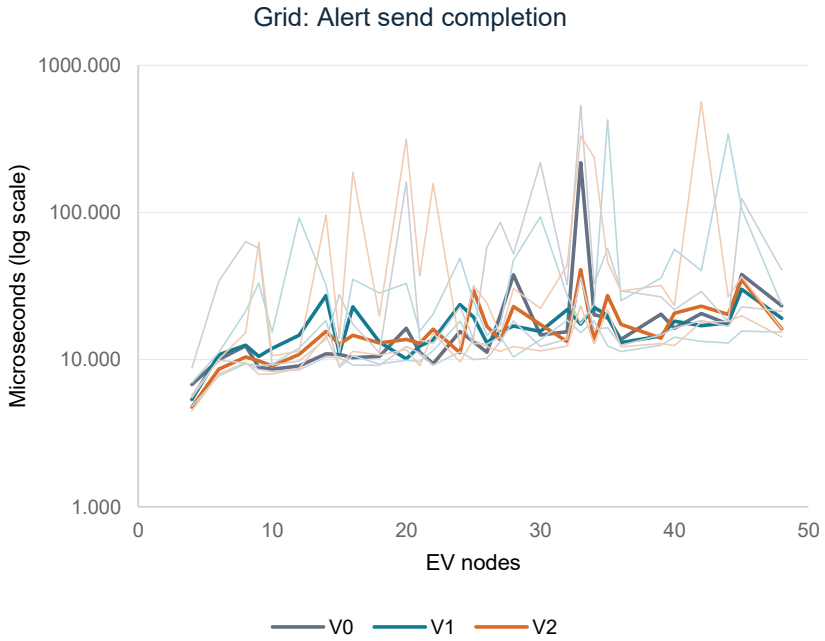
Neighbour reply phase

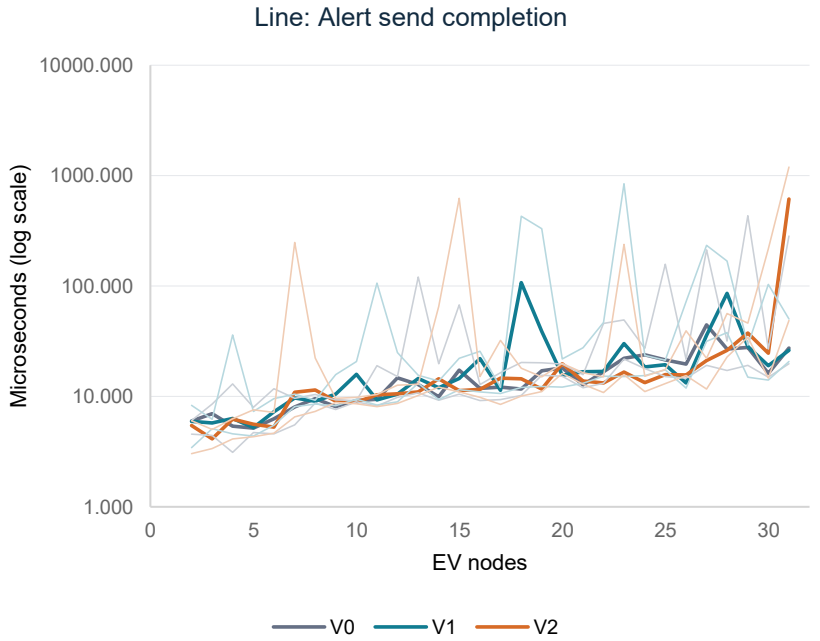




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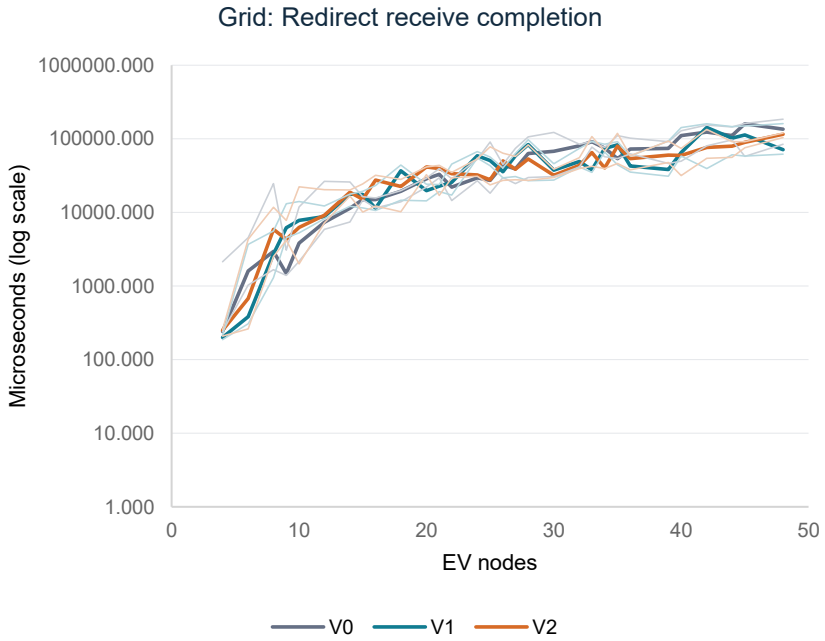
Alert send completion

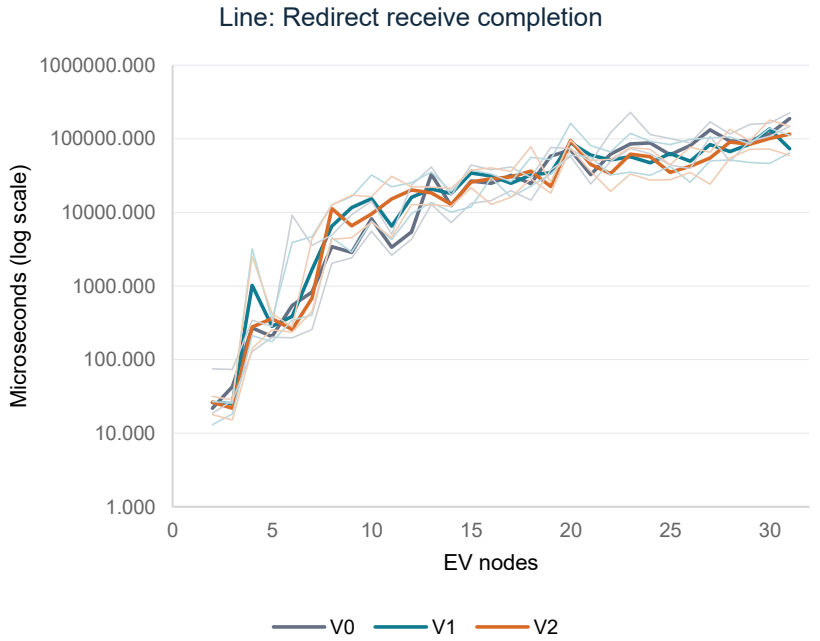




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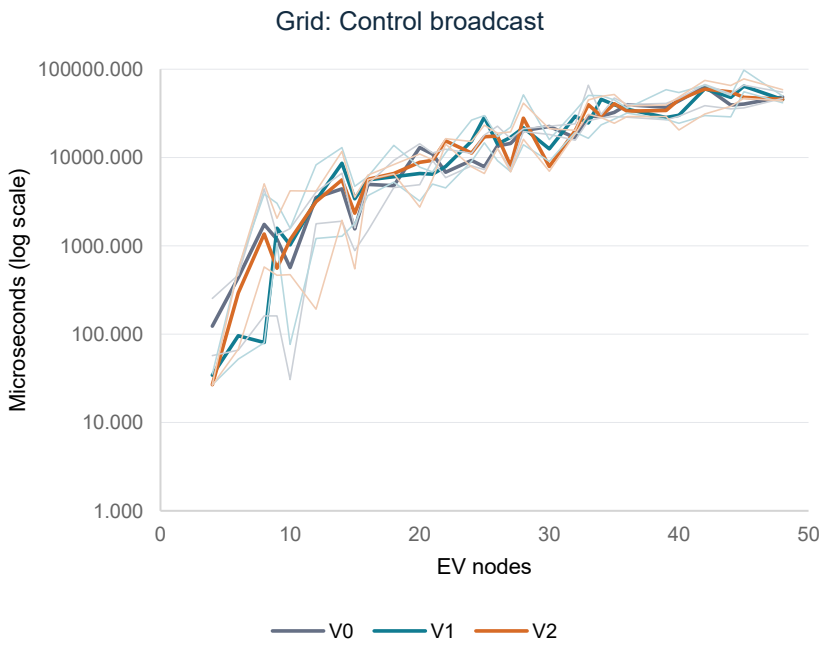
Redirect receive completion

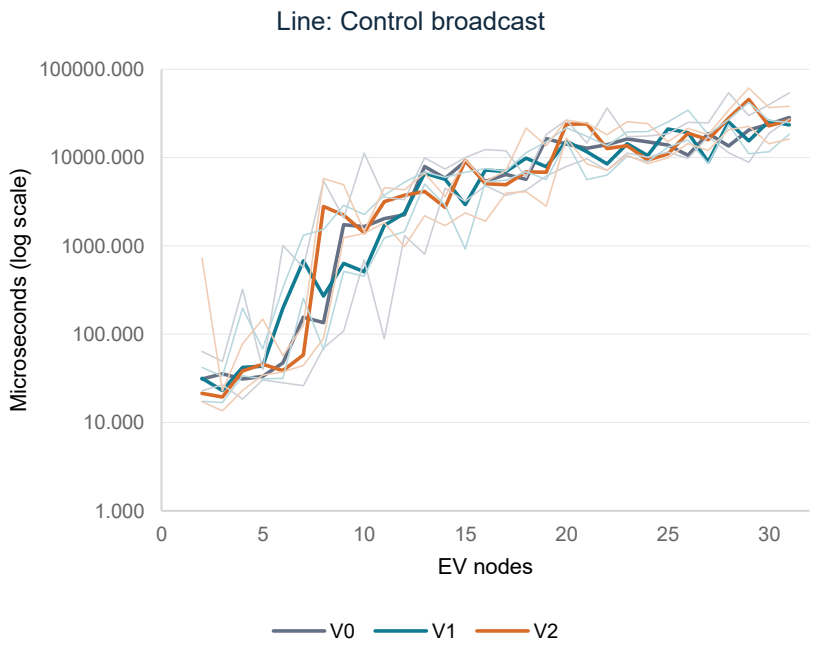




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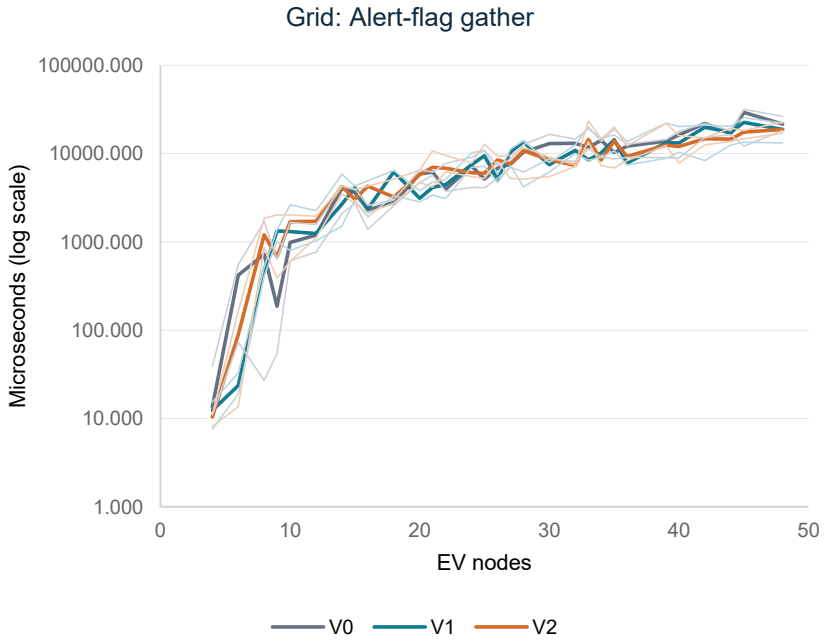
Control broadcast

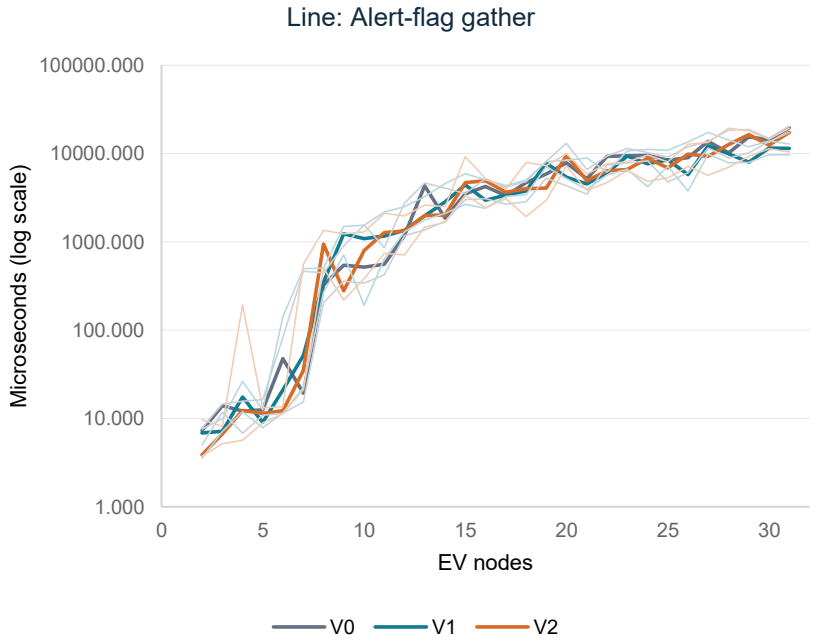




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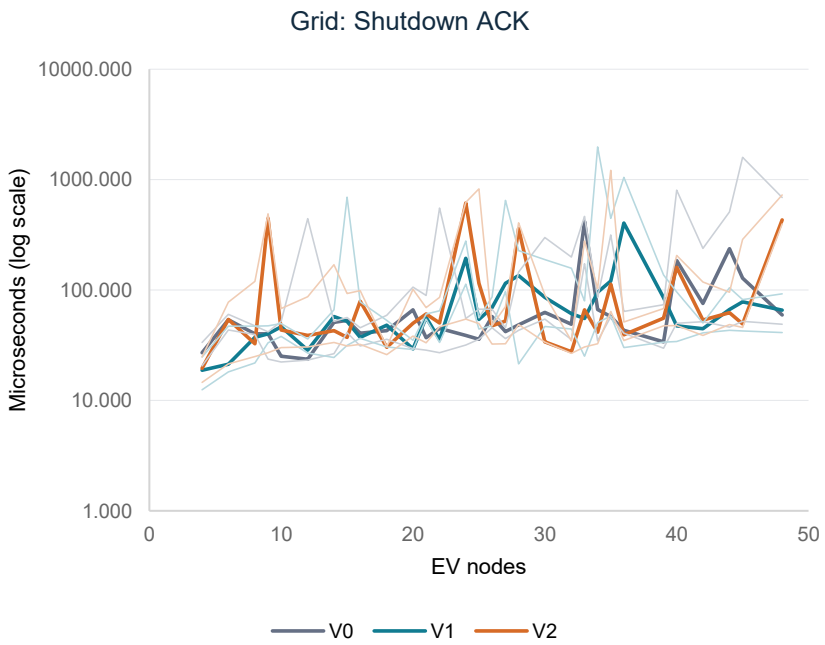
Alert-flag gather

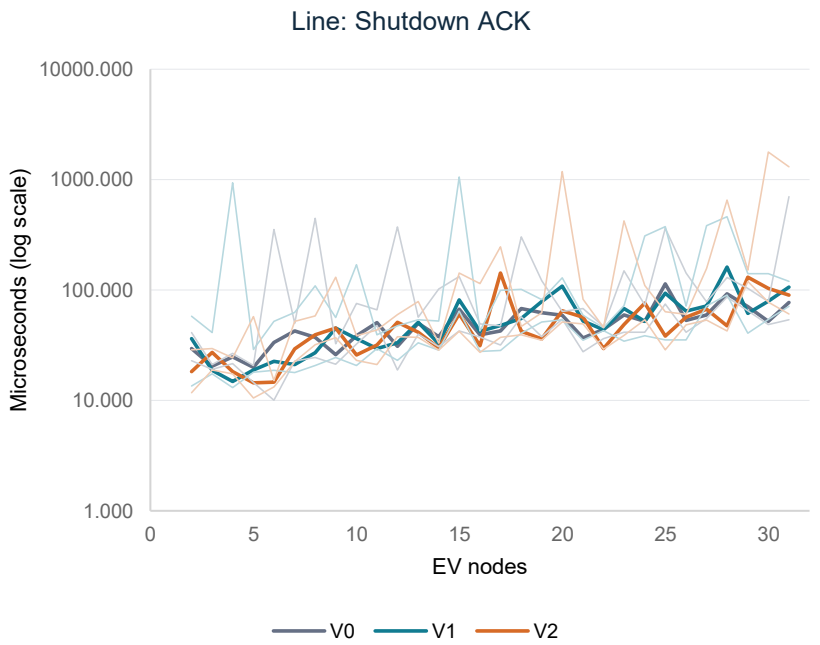




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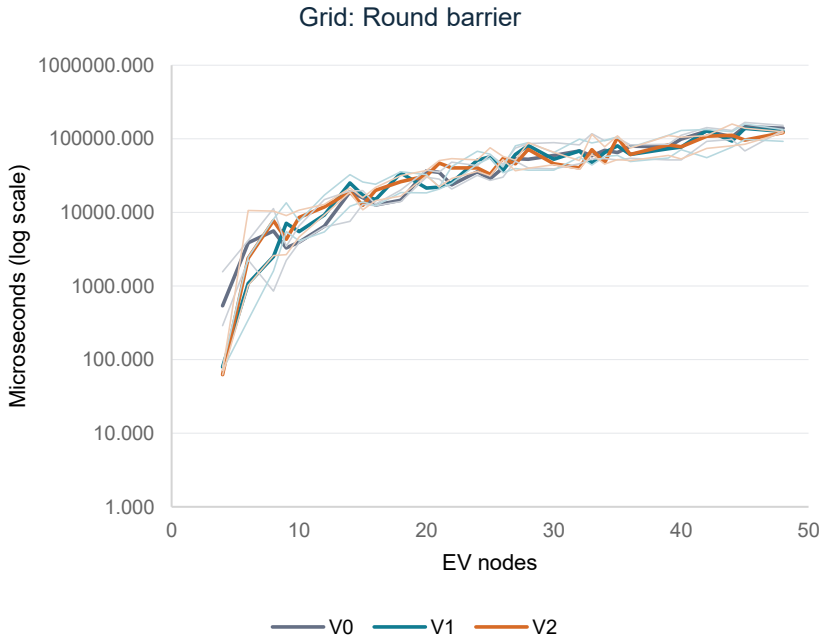
Shutdown ACK

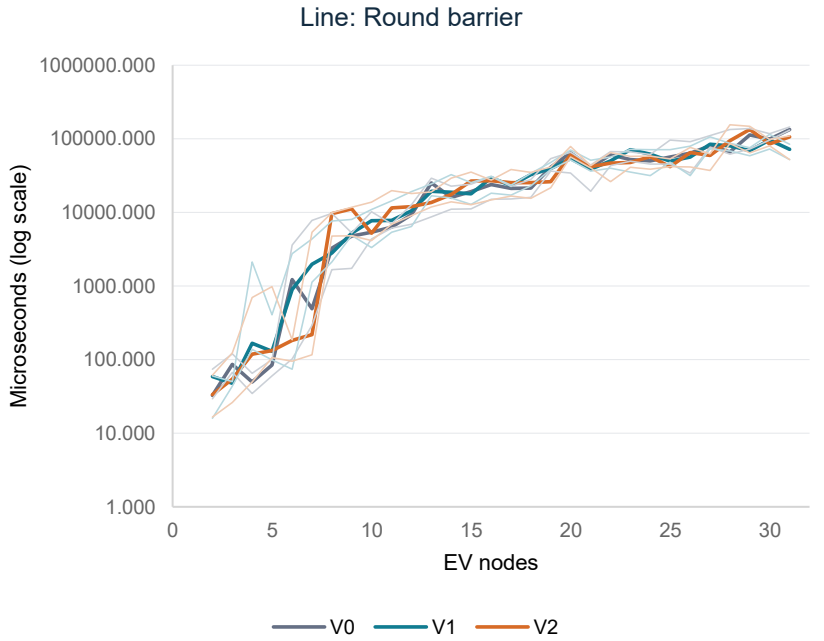




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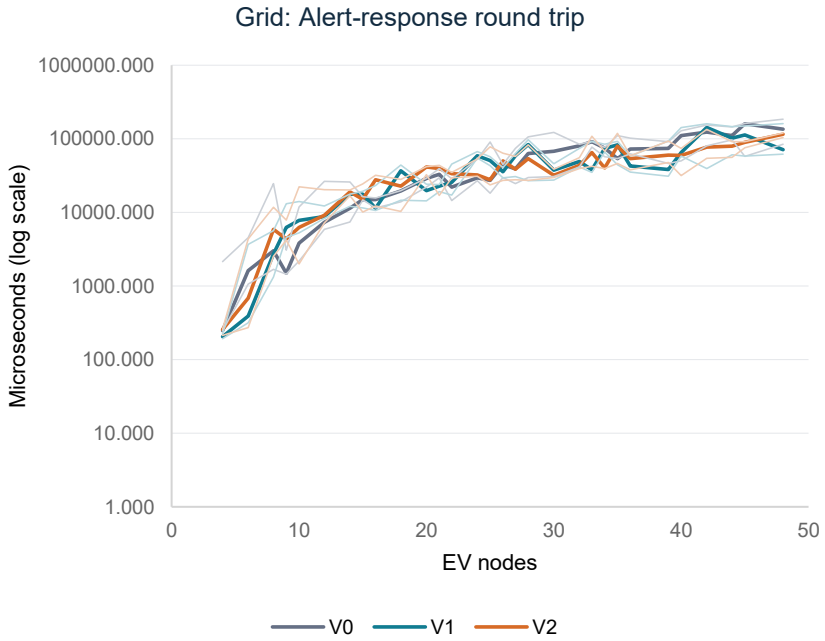
Round synchronization

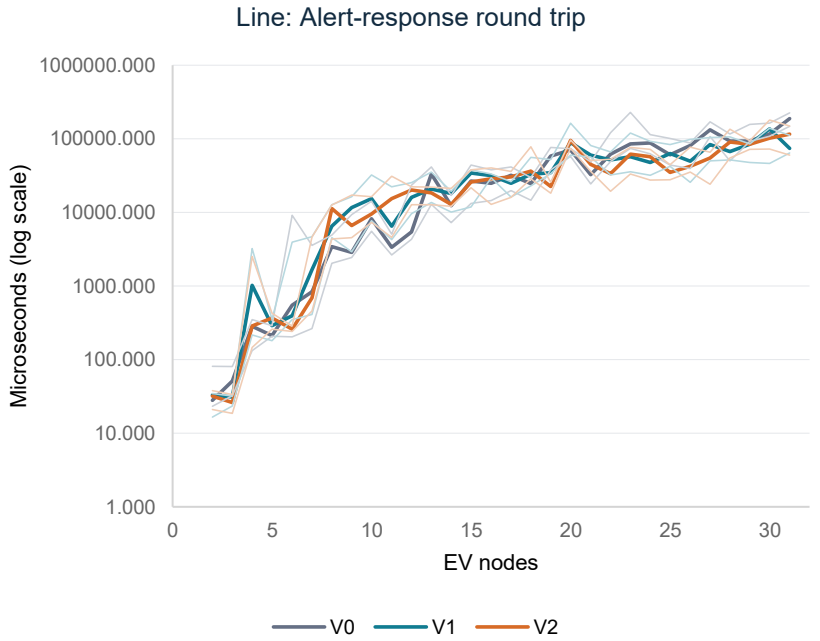




Every measured size is retained. Lines show medians of three independent run means. Pale boundary lines show the observed minimum and maximum.

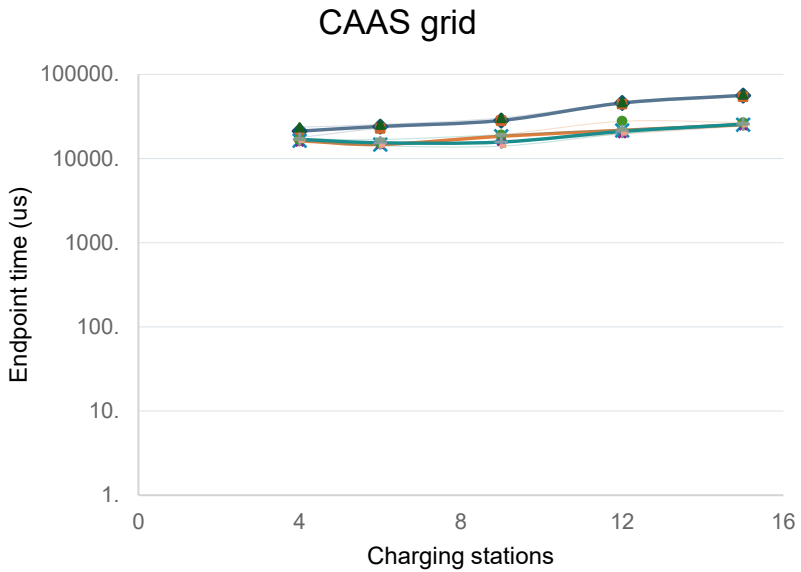
Alert-response round trip

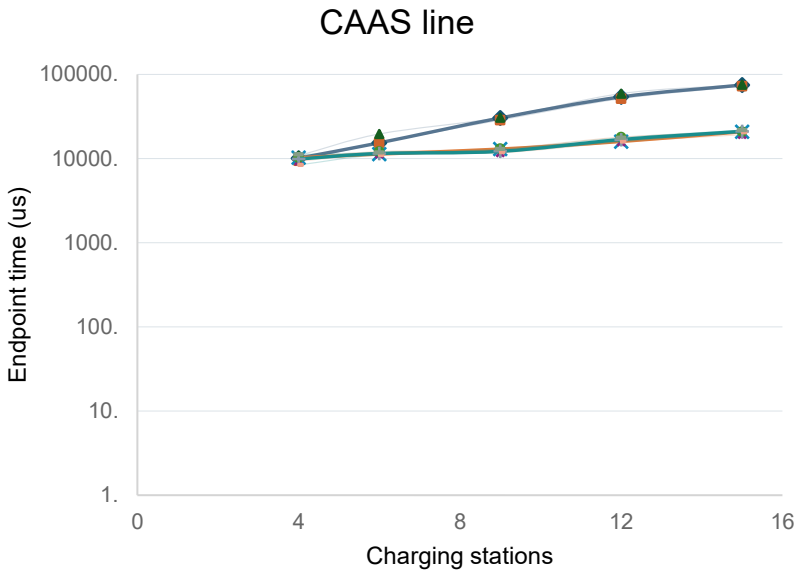




Every measured size is retained. Lines show medians of three independent run means. Pale boundary lines show the observed minimum and maximum.

CAAS: Neighbour query phase

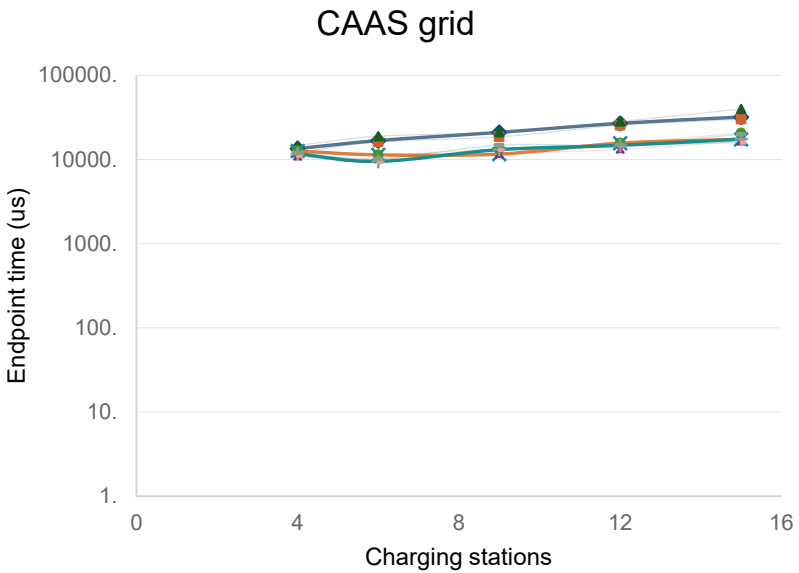


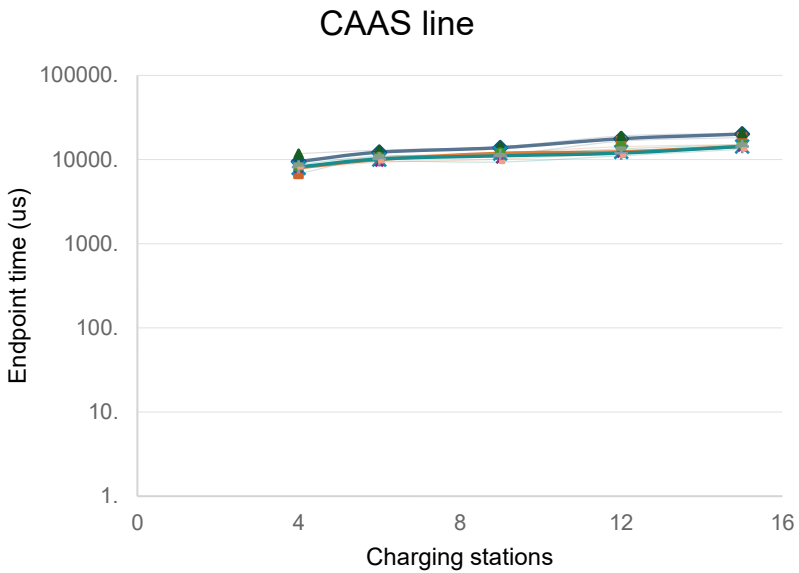


V0: blue-grey V1: orange V2: teal. Medians of 3 launches; pale lines are min/max.

Per measured cycle. Five sizes; log y-axis. Station counts exclude the base. All raw data are retained.

CAAS: Neighbour reply phase

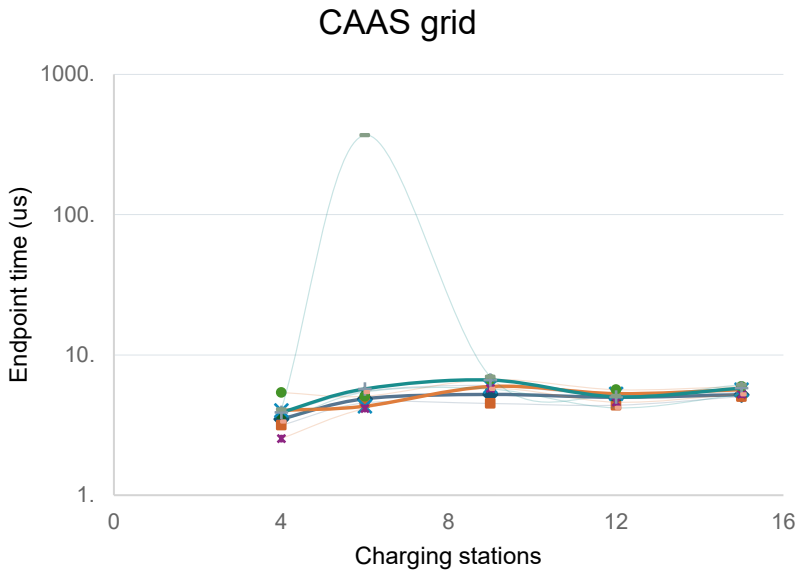


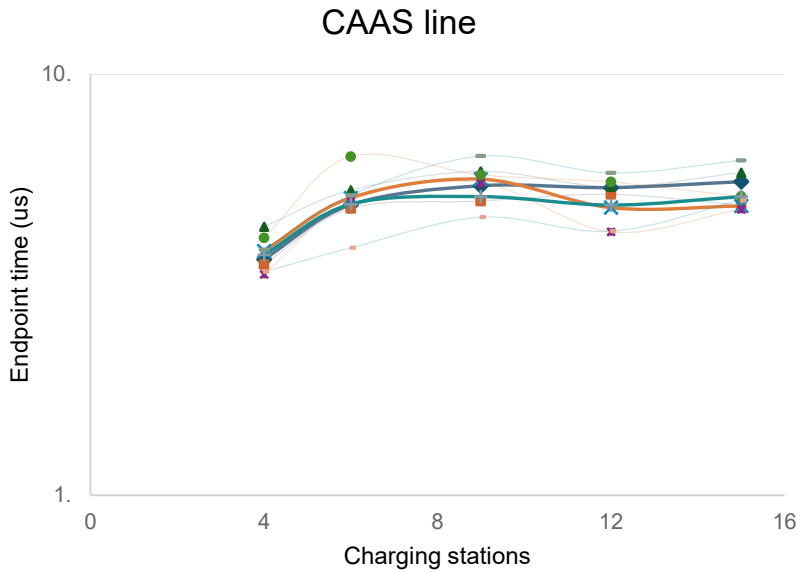


V0: blue-grey V1: orange V2: teal. Medians of 3 launches; pale lines are min/max.

Per measured cycle. Five sizes; log y-axis. Station counts exclude the base. All raw data are retained.

CAAS: Alert send completion

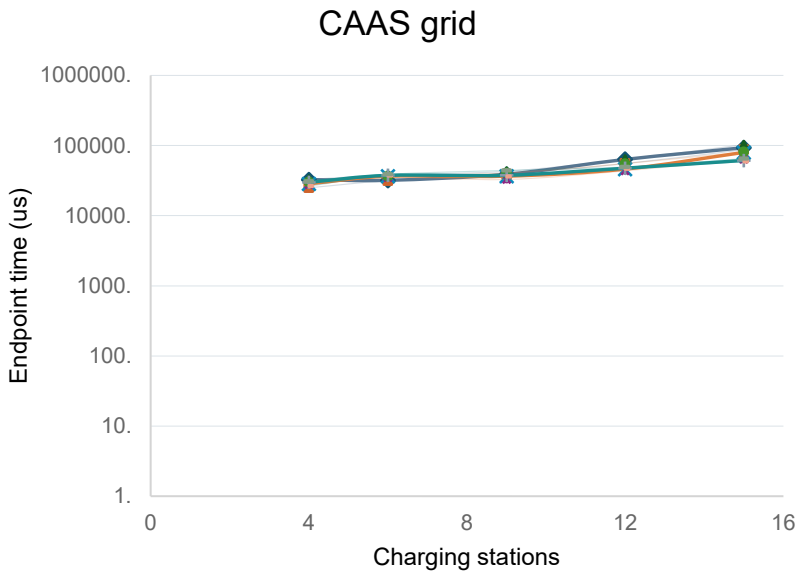


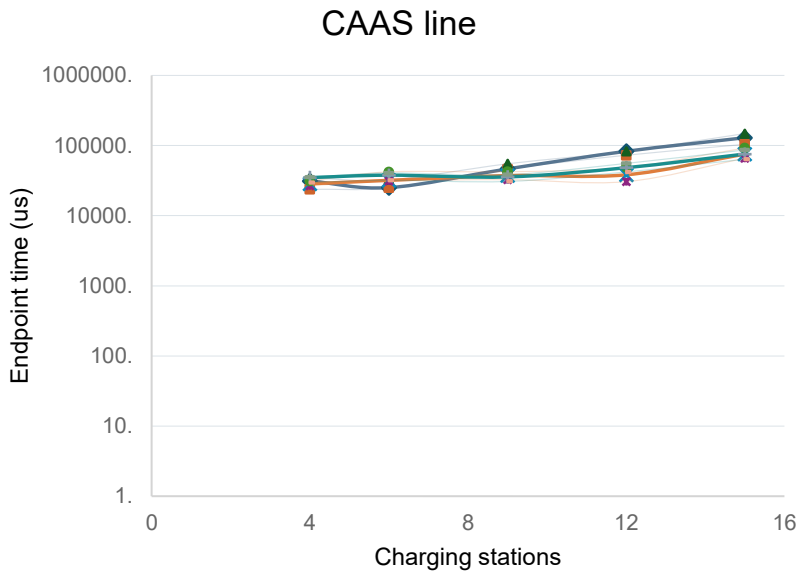


V0: blue-grey V1: orange V2: teal. Medians of 3 launches; pale lines are min/max.

Per completed operation. Five sizes; log y-axis. Station counts exclude the base. All raw data are retained.

CAAS: Redirect receive completion

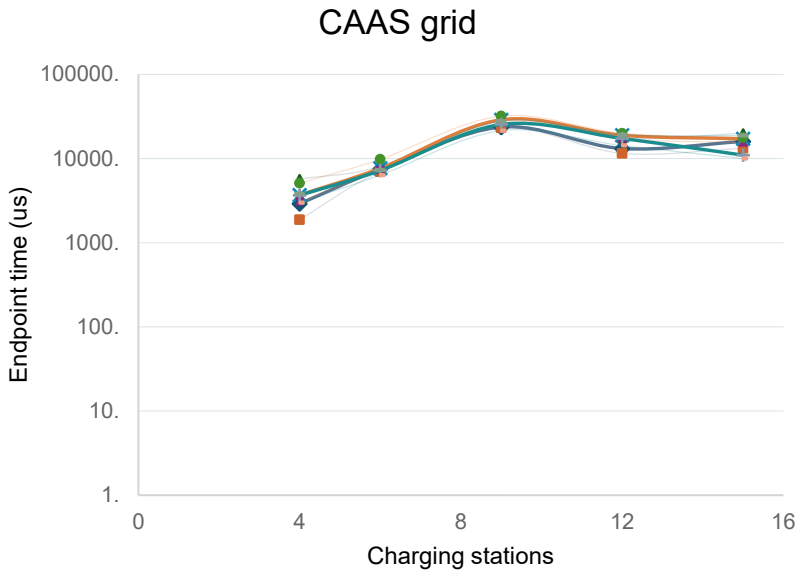


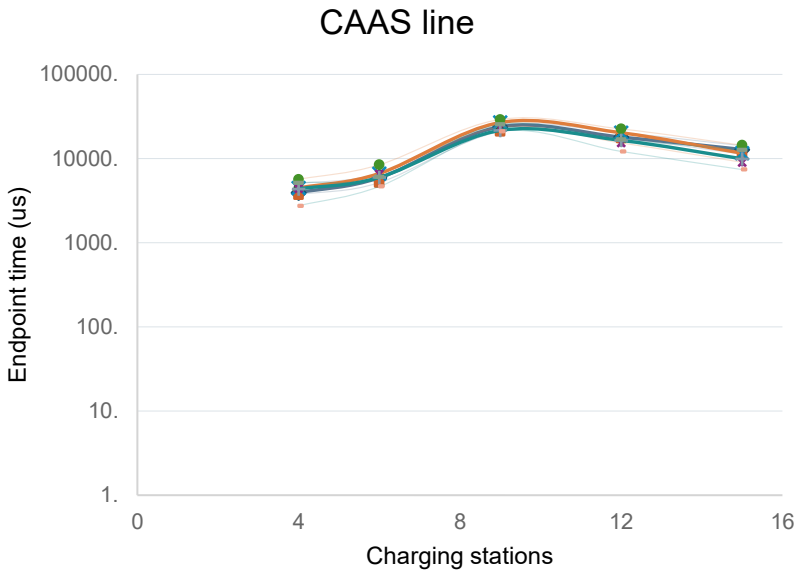


V0: blue-grey V1: orange V2: teal. Medians of 3 launches; pale lines are min/max.

Per completed operation. Five sizes; log y-axis. Station counts exclude the base. All raw data are retained.

CAAS: Control broadcast

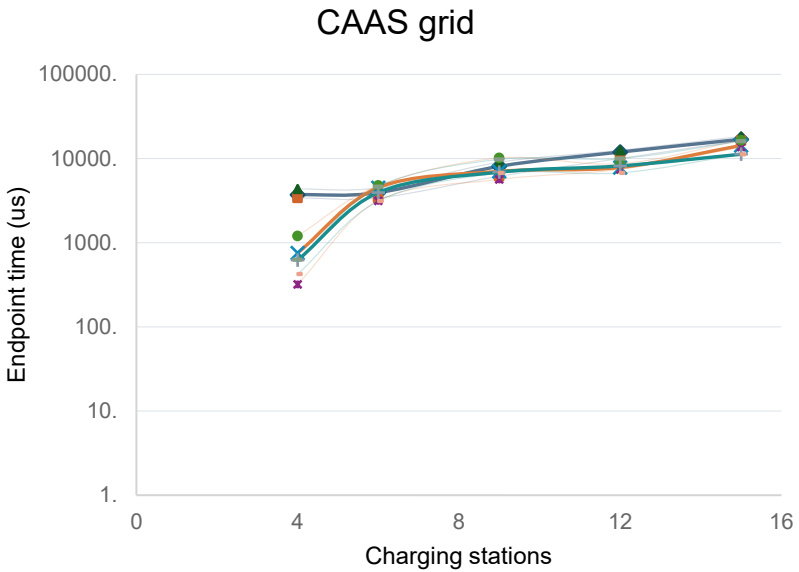


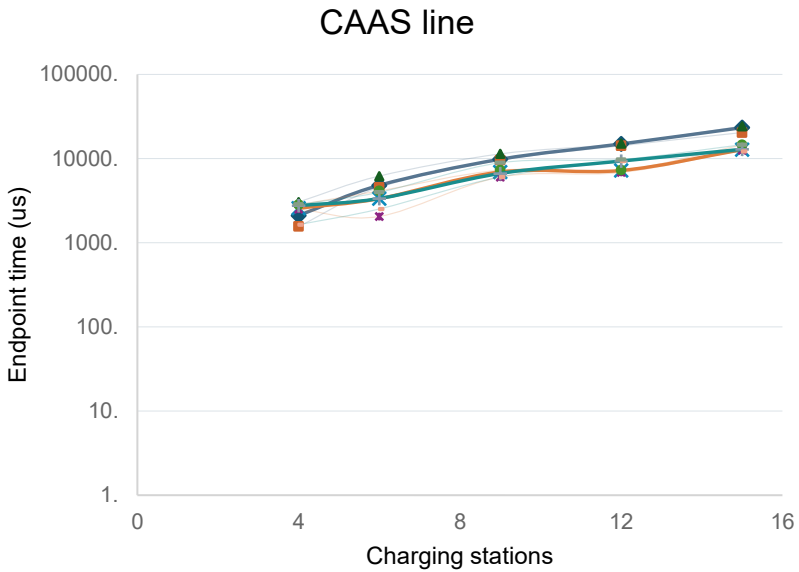


V0: blue-grey V1: orange V2: teal. Medians of 3 launches; pale lines are min/max.

Per completed operation. Five sizes; log y-axis. Station counts exclude the base. All raw data are retained.

CAAS: Alert-flag gather

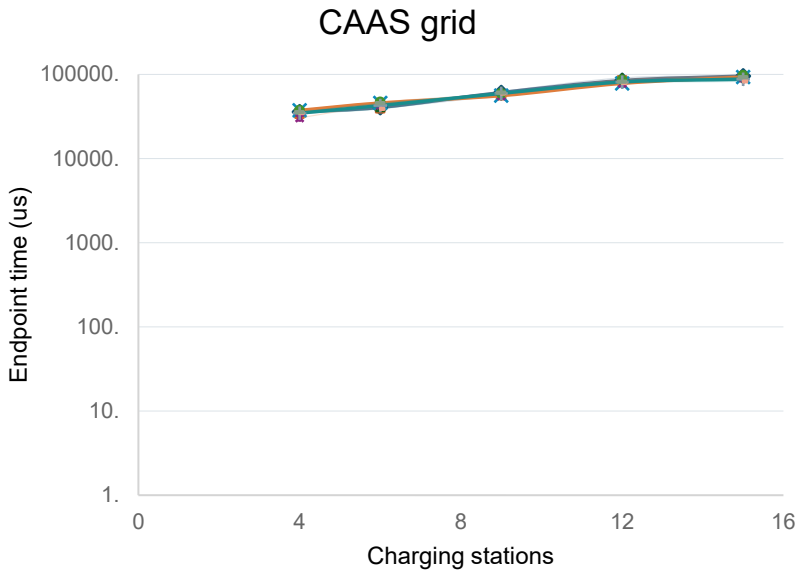


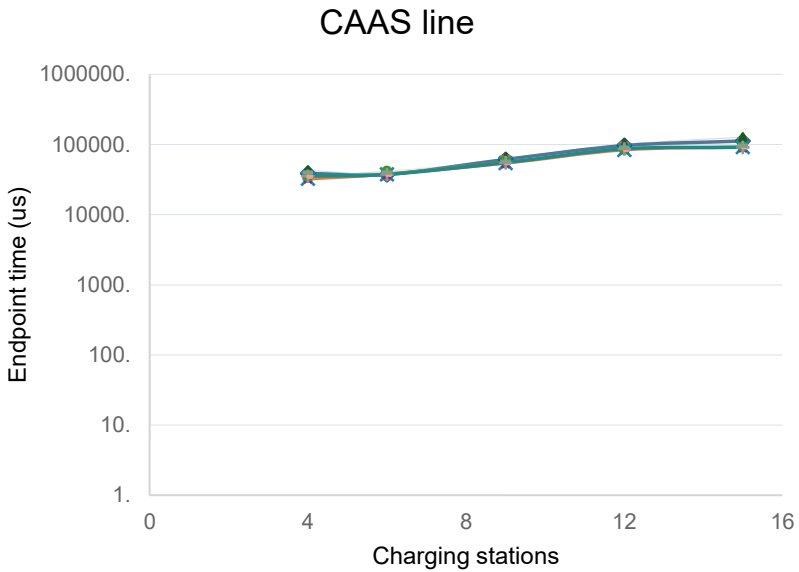


V0: blue-grey V1: orange V2: teal. Medians of 3 launches; pale lines are min/max.

Per completed operation. Five sizes; log y-axis. Station counts exclude the base. All raw data are retained.

CAAS: Round synchronization

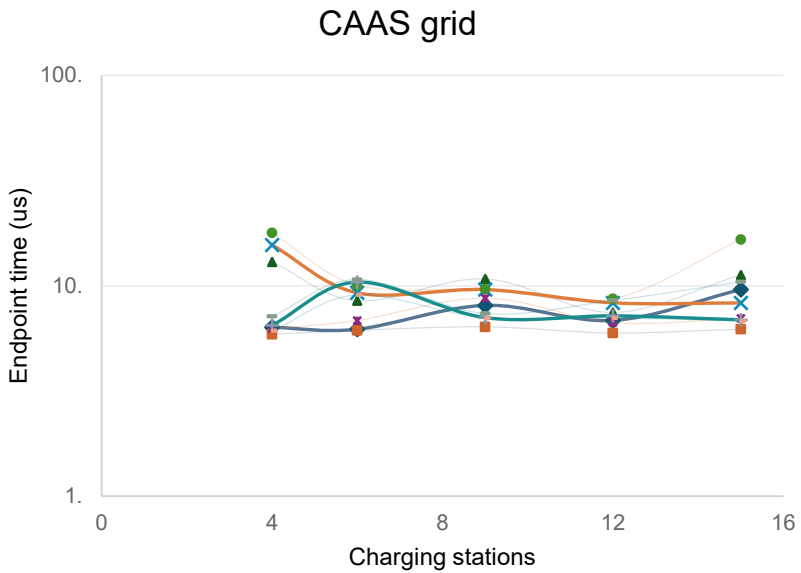


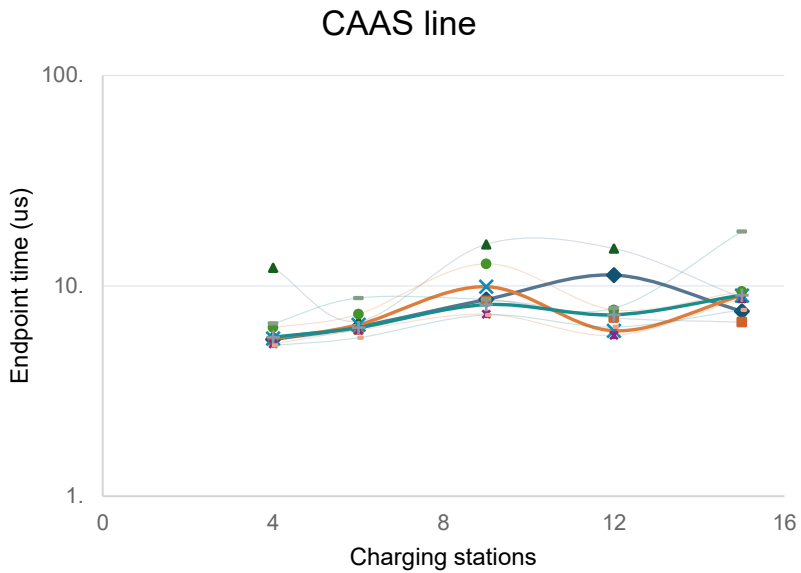


V0: blue-grey V1: orange V2: teal. Medians of 3 launches; pale lines are min/max.

Per completed operation. Five sizes; log y-axis. Station counts exclude the base. All raw data are retained.

CAAS: Shutdown ACK

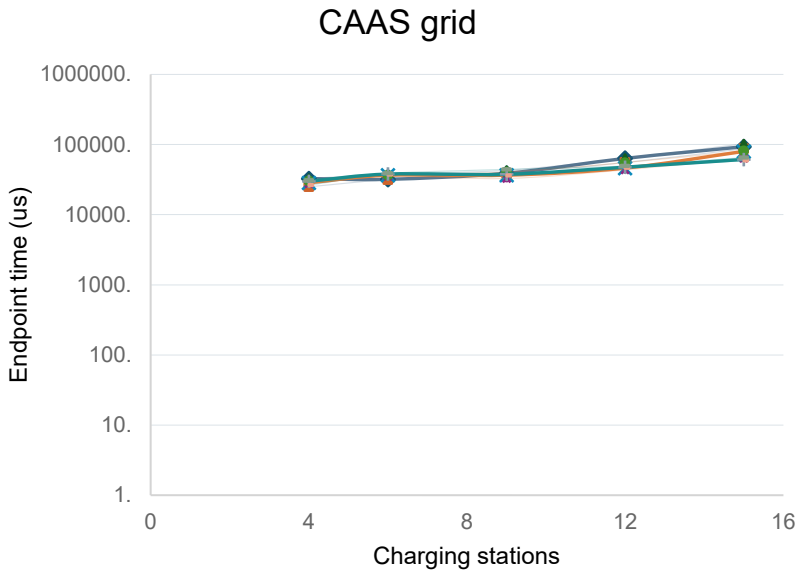


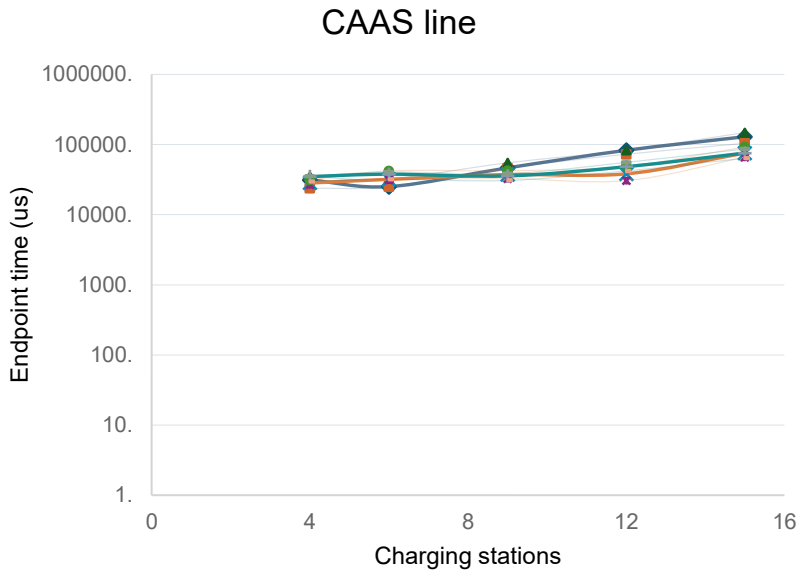


V0: blue-grey V1: orange V2: teal. Medians of 3 launches; pale lines are min/max.

Per completed operation. Five sizes; log y-axis. Station counts exclude the base. All raw data are retained.

CAAS: Alert-response round trip





V0: blue-grey V1: orange V2: teal. Medians of 3 launches; pale lines are min/max.

Per completed operation. Five sizes; log y-axis. Station counts exclude the base. All raw data are retained.

CAAS completion and correctness evidence

CAAS job 199603: execution and completion

Local read-only evidence viewer. Content below comes from downloaded CAAS files; this is a browser view of those records.

SLURM accounting: selected original rows

```
JobID|JobName|State|ExitCode|Elapsed|NodeList|AllocCPUS
199603|fit3143-lab3-audit|COMPLETED|0:0|00:06:54|student-caas-n[10-11]|32
199603.batch|batch|COMPLETED|0:0|00:06:54|student-caas-n10|16
```

Correctness and completion: selected original stdout lines

```
PASS: ties, recent alert, recent low observation, expiry, isolated node
PASS: independent oracle for every station in all 1..7 by 1..7 shapes, four cache states, three variants
PASS 3x3 ties: all three versions; complete shutdown
PASS: 24 correctness runs, 90 scaling runs, 4 expiry runs, 1 realistic run, candidate tests and benchmark.
SCALING 90/90 line_1x15_v1_r2 4.949s
CAAS_EXPERIMENTS_COMPLETE
```

Source files

```
results/caas/job_accounting.txt
results/caas/slurm-199603.out
Full source logs and per-rank raw timings are retained in the submission package.
```

Authentic screenshot of the local evidence viewer displaying unmodified CAAS scheduler/test output.

CAAS all-ties and shutdown evidence

CAAS 3x3: all nearest ties and graceful shutdown

Local read-only evidence viewer. Content below comes from downloaded CAAS files; this is a browser view of those records.

Actual recorded launch argv

```
srun --mpi=none --exact --nodes=2 --ntasks=10 --ntasks-per-node=5 --cpus-per-task=2 --distribution=cyclic --cpu-bind=cores ./wsn_caas 3 3 2 --
variant 2 --scenario ties --rounds 4 --warmup 0 --tick 0 --expiry 30 --output results/caas/check_3x3_ties_v2
```

Original base log excerpt: header, first incident and shutdown summaries

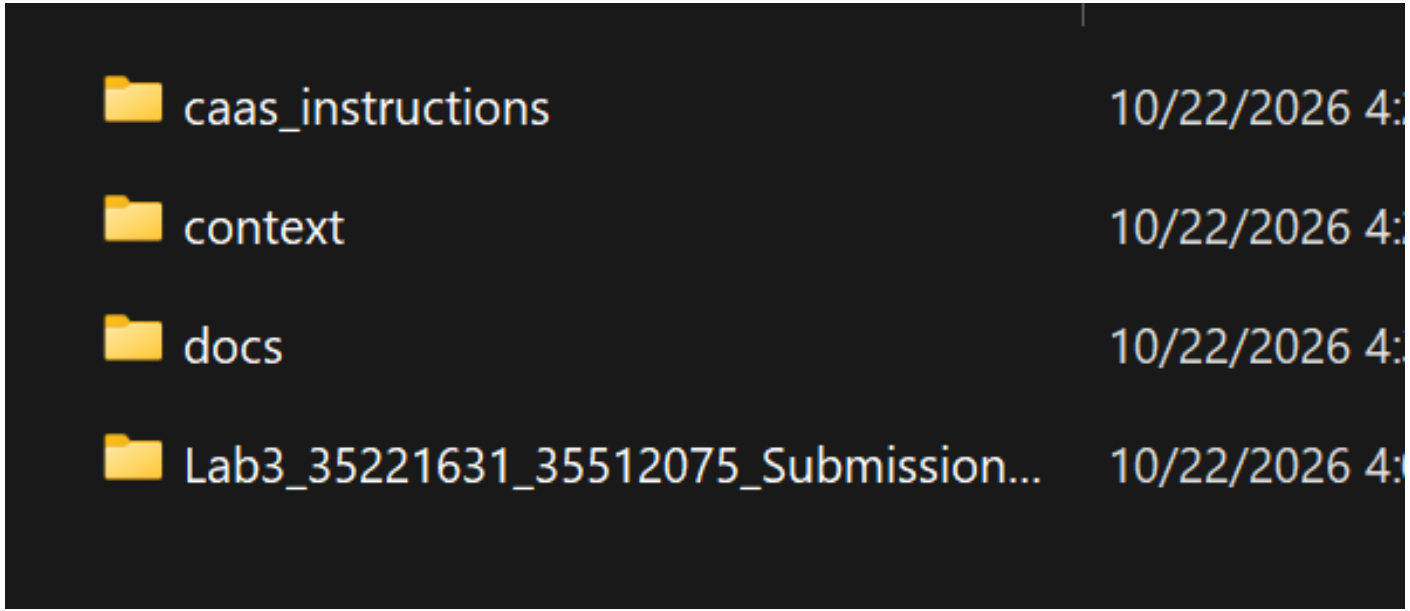
```
# host=student-caas-n10 world=0 station=-1 variant=2 ports=2 threshold=1 expiry=30 tick=0
{"round":0,"source":0,"coordinates":[0,0],"discovered_utc":"2026-10-
22T10:18:29Z","free":0,"neighbour_messages":8,"neighbour_bytes":32,"neighbours":[{"rank":-2,"free":-1},{"rank":3,"free":0},
{"rank":-2,"free":-1},{"rank":1,"free":0}],"inferred_candidates":[2,4,6]}
# alerts=4 elapsed_seconds=0.781598678 shutdown_acks=9
# measured_point_to_point_messages=145 measured_payload_bytes=900 collective_operations=13 (warmup excluded; MPI protocol overhead excluded)
```

Source record

```
CAAS job 199603
Raw file: check_3x3_ties_v2.base0.log
The complete incident sequence is retained in all_base_station_logs.log.
```

Authentic screenshot of the local evidence viewer displaying a CAAS 3x3 all-ties base log.

Supplied project folder structure



User-supplied local folder structure: the assessment inputs and original implementation package.

This image records file organisation only. CAAS execution is evidenced by the separate verified logs and captures.

Station architecture and simulation settings

Ranks and topology

Exactly rows × columns + 1 ranks. MPI_Comm_split excludes world rank 0 from the worker grid. MPI_Cart_create disables reordering and wraparound. MPI_Cart_shift returns MPI_PROC_NULL at missing edges.

Port ownership and synchronization

ports[k]: 1 = occupied, 0 = free. Each persistent thread owns one element and private xorshift state. An atomic sum counts free ports. MPI_THREAD_FUNNELED restricts MPI calls to the initial thread.

Defaults and workload

Rows, columns and ports are required. Defaults: threshold 1, tick 10 s, expiry 30 s, 10 rounds, seed 3143. Random mode uses 0.75 occupancy probability per port per update. Full, empty, ties and mixed modes support checks.

Matching phases and report contents

Deterministic neighbour matching

Each real edge exchanges one integer intent in both directions, including zero intentions. Capacity replies go only to requesting neighbours. A congested station selects a suitable immediate neighbour with the lowest rank.

Batched alerts and advice

After the alert-flag gather, each alert sends one report. The base receives the whole batch, updates every cache entry, then returns a count and all surviving distance-two candidate ranks in rank order. An empty list is valid.

Report and end of cycle

Fields: station rank, coordinates, cycle, UTC discovery time, local/neighbour capacities, message counts and payload bytes. The station logs advice for simulated vehicles, waits for its tick and joins the global round barrier.

Application work and message counts

Operation	Cost or count
Port snapshot	O(k) work/storage per station; O(1) per thread + synchronization
Grid edges	$E = m(n - 1) + n(m - 1)$
Neighbour exchange / round	2E intents and at most 2E capacity replies
Base exchange / round	2A report/advice messages for A alerts
Candidate search, V0	O(N) per alert; O(A × N) per cycle
Candidate search, V1/V2	At most 8 offsets: O(1) per alert; O(A) per cycle
Storage and shutdown	O(N) base caches/batches; N shutdown ACKs

N stations, k ports, m × n grid. Degree ≤4 (≤2 in a line). At most 8 tied candidates, or 2 in a line. Counts exclude MPI internal collective packets.

Timing definitions and normalization

Metric	Measured interval	Divide station total by
query / reply	Neighbour phase elapsed time	Measured rounds
report	Station MPI_Send completion	Report sends
redirect	Advice receive wait, including base queueing	Advice receives
control	CONTINUE + final STOP broadcast	Measured rounds + 1
flags	Alert-flag gather elapsed time	Measured rounds
sync	Global round barrier elapsed time	Measured rounds
ack	Station shutdown send completion	1 per station per launch
alert_roundtrip	Report send through advice receipt	Alerts in measured rounds

Every interval uses MPI_Wtime on the same rank. Send completion can reflect eager buffering. These are endpoint operation costs and include waiting.

Aggregation and timer boundaries

From raw records to graph points

Normalize each active station by the metric-specific denominator, then average stations equally within each launch. Plot the median of 3 launch means, with observed min/max. Base-rank totals remain in the raw CSV.

Logical messages and bytes

Query/reply counts are outgoing application messages, excluding MPI_PROC_NULL, rather than every Irecv or Waitall API call. Payload bytes exclude MPI protocol overhead. ACK has one observation per station per launch.

What the timers include

Station operation timers do not directly wrap sleep, startup or communicator setup. The station barrier follows its tick but captures imbalance. The base barrier can wait for ticks. Simulation elapsed time includes STOP/ACKs.

Experimental controls

Control	Local study	CAAS study
Coverage	30 grids + 30 lines; 540 scaling runs	5 grids + 5 lines; 90 scaling runs
Variants / repeats	V0, V1, V2; 3 launches each	V0, V1, V2; 3 launches each
Cycles / workload	2 warmup + 6 measured; 2 ports; mixed	2 warmup + 12 measured; 2 ports; mixed
Resources	2 TCP containers; shared 16-CPU WSL host	2 allocated compute nodes; 16 rank slots
Placement / toolchain	Explicit oversubscription; no link delay	2 cores/rank; bound cores; passive OMP waits
CAAS sizes	Local full coverage retained separately	Grids: 2×2, 2×3, 3×3, 3×4, 3×5 Lines: 4, 6, 9, 12, 15

Variant order rotates between repeats. CAAS uses Open MPI 4.1.5 / GCC 11.2.0. Different cycle counts and resources prevent a controlled cross-environment hardware speedup claim.

Local implementation comparison

Topology / metric	Median V0/V1	Median V0/V2
grid / query	2.581	2.272
grid / report	0.938	0.898
grid / redirect	1.086	0.987
grid / alert_roundtrip	1.095	0.986
line / query	3.024	2.762
line / report	0.982	1.022
line / redirect	0.844	1.042
line / alert_roundtrip	0.850	1.042

Ratios >1 favour the refinement. Across 60 matched sizes, round-trip scores: V0 1.000, V1 0.987, V2 1.018. The V2 preference is marginal and descriptive.

Local scaling summary for V2

V2 metric	Grid first / last 5 (µs)	Ratio	Line first / last 5 (µs)	Ratio
query	3312.21 / 52526.58	15.86	27.72 / 28466.81	1026.80
reply	1878.64 / 28205.51	15.01	19.88 / 18665.77	938.95
report	9.05 / 20.66	2.28	5.43 / 26.21	4.82
redirect	4336.81 / 79396.58	18.31	255.48 / 90396.17	353.82
control	557.77 / 48136.42	86.30	38.45 / 26639.90	692.83
flags	678.06 / 14698.84	21.68	11.46 / 12613.40	1100.52
sync	4314.61 / 108662.00	25.18	118.54 / 94030.18	793.23
ack	43.41 / 62.32	1.44	18.20 / 89.93	4.94
alert_roundtrip	4346.91 / 79424.81	18.27	260.26 / 90419.46	347.42

First/last five size summaries. Degree stays bounded while total traffic grows O(N). Scheduling, synchronization and base fan-in affect latency. A line message does not traverse N forwarding hops.

CAAS implementation comparison

Metric	Grid V0/V1	Grid V0/V2	Line V0/V1	Line V0/V2
query	1.639	1.806	2.341	2.492
reply	1.720	1.769	1.220	1.250
report	0.921	0.897	0.964	1.061
redirect	1.148	1.093	1.246	1.302
control	0.819	0.932	0.889	1.095
flags	1.184	1.460	1.440	1.476
sync	1.029	1.039	1.148	1.110
ack	0.825	0.978	0.978	1.010
alert_roundtrip	1.148	1.093	1.246	1.302

Round-trip scores over 10 matched sizes: V0 1.000, V1 1.157, V2 1.197. Three launches provide descriptive evidence. Results do not establish a universal best variant.

Isolated CAAS candidate benchmark

N stations	V0 seconds	V1 seconds	V2 seconds	Checksum
10	0.000424762	0.000232499	0.000237990	16000
100	0.004091474	0.000240133	0.000229484	19600
1000	0.039446566	0.000234573	0.000229534	19960
10000	0.392703870	0.000229474	0.000229663	19996

10,000 actual calls per size/version. All checksums match independent expected values. At N=10,000, V0/V2 = 1709.9× for candidate computation only. MPI exchange and port-thread costs are excluded.

Correctness checks and validation

Build and integration checks

Strict C11/OpenMP build: -Wall -Wextra -Wpedantic -Werror. 25 local runs cover isolated, line, rectangular, full, empty, all-ties and random cases, custom thresholds/expiry, and invalid inputs/rank counts.

Independent candidate oracle

Enumerates every rank in every 1..7 by 1..7 shape, four cache states and all versions. Checks boundaries, all eight offsets, expiry and recent low observations. Address/undefined-behaviour sanitizers passed.

CAAS verification

24 correctness simulations plus candidate tests. Deterministic decisions agree across variants. Validation checks source equality, successful exits, all metric rows, message/byte totals, report fields, two hosts and every shutdown ACK.

Source and result traceability

Final measured source

src/wsn.c matches the CAAS environment record and measured snapshot. SHA-256: 28cd1d3951e9b4fa523765811bff2f3da3c747916e08a66d0d4e665b562fdcdd

Changes since the local measured snapshot

Added identification comments and a post-measurement summary. Corrected a flush/close comment. Captured UTC discovery time before the alert gather. Rejected out-of-range ticks before conversion to the system time type.

Separate evidence chains

Local results remain tied to results/cluster/timed_source_snapshot.c. CAAS measures the final source. Candidate decisions and communication methods remain unchanged. Analysis scripts reproduce aggregates from retained raw records.

Normal shutdown and MPI completion

STOP and thread-team exit

After the configured cycles, the base broadcasts STOP. Port teams observe it at a barrier and exit. Every nonblocking request batch completes before its buffers are released. Waitall permits safe reuse without proving useful overlap.

Acknowledgements and cleanup

Stations flush logs, send ACKs and free port/PRNG arrays. The base receives all N ACKs and closes its log. The final metrics gather completes, files close, communicators are freed and all ranks call MPI_Finalize.

Normal and error paths

Normal shutdown uses the protocol rather than Ctrl+C or kill. Invalid arguments, allocation failures and I/O failures use MPI_Abort. Preposted receives establish matching without depending on eager buffering.

CAAS execution and verified recovery

Successful complete study

Job 199603: COMPLETED, exit 0:0, on student-caas-n10 and student-caas-n11. 119 simulations, 10,440 raw metrics and 1,041 station placements. All station ACKs are present.

Launch route and placement

srun --mpi=none --exact, explicit rank counts and two bound cores/rank. 60 launches emitted the known uneven-task cap warning. Checked placement stays balanced, with at most 8 ranks per node and no unexpected resource errors.

Excluded preparation attempts

199444: forced-PMIx timeout. 199460: daemon callback failure. 199513: none/pmi2 probes succeeded, pmix/mpirun timed out. 199520 exceeded 5 minutes. Final job used 10 minutes. Failed/probe timings are excluded.

Limits and future improvements

Model and routing limits

Synthetic port arrivals omit real EV traces, radio interference, packet loss and road geometry. The fixed distance-two frontier does not expand automatically. Inferred availability needs confirmation or reservation in a deployed service.

Scaling and portability

Global gather/barrier phases and one base create coordination and fan-in costs. Reports use MPI_BYTE with native C structs, assuming homogeneous layouts. Extensions need portable serialization and improved progress/termination design.

What the measurements support

Local containers share an oversubscribed host. CAAS verifies distinct allocated hosts, not independent underlying hardware. Three launches and small CAAS size sets limit generalization. Longer controlled runs and realistic workloads remain next steps.